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Review article

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Digital twins and digital models of the human circulatory system

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Abstract

Digital models and digital twins of human circulatory transport could transform the way cardiovascular and haematological diseases are understood, monitored and treated. Digital twins are dynamic virtual representations of physical systems that continuously assimilate real-world data to simulate and predict system behaviour. However, translating digital twins into clinical practice remains challenging owing to the complexity of human physiology and the need for continuous bidirectional coupling between virtual models and their physical counterparts. Advances in medical-grade sensors, wearable devices, microfluidics, artificial intelligence and high-performance computing are accelerating the evolution of digital models into clinically meaningful digital twins. In this Review, we examine how digital twins can model the human circulatory system across scales, from macroscopic blood flow to molecular and cellular transport. We outline the essential components of a circulatory-transport digital twin, describe the pathophysiological conditions that can be digitally represented, and discuss approaches for acquiring and integrating physiological data, computational modelling strategies and model-based inference. We further survey applications of digital models and digital twins across various types of model inferences, from mechanistic insights to clinical decisions such as disease diagnosis, risk stratification, surgical planning and treatment planning. Finally, we identify key challenges and opportunities for next-generation circulatory digital twins capable of real-time monitoring, predictive simulation and closed-loop therapeutic control.

Sections

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Key points

- Digital models, digital shadows and bidirectional digital twins form a continuum, distinguished by increasing levels of physiological integration, data assimilation and clinical decision support.
- Circulatory digital twins provide a multiscale representation of transport, linking organ-level haemodynamics with vascular adaptation and cellular biomechanics across cardiovascular and haematological disease.
- Hybrid modelling strategies that couple physics-based solvers, artificial intelligence and high-performance computing enable scalable, mechanistically consistent personalization of circulatory dynamics prediction.
- Clinical translation of circulatory digital twins will depend on data integration, validation and uncertainty quantification, as well as scalable learning paradigms, including operator-learning and foundation models to support longitudinal monitoring and treatment planning.

Introduction

The concept of a digital twin originated from the need for accurate virtual representations of physical entities that can inform design, monitoring and optimization processes[1]. Within this framework, virtual representations are typically conceptualized along a continuum, from digital models to digital shadows and, ultimately, digital twins (Fig. 1a). A digital model is a static computational representation without continuous data exchange with the physical system. A digital shadow incorporates a one-way data stream from the physical entity to the virtual model, enabling the latter to reflect real-time changes in the physical counterpart. By contrast, a fully realized digital twin features bidirectional data exchange, allowing the virtual entity not only to mirror, but also to influence or control the physical system. Guided by the 'fit-for-purpose' principle, in which the scope and parameterization of the model are tailored to the decision task and computational constraints, digital twins enable process optimization, predictive modelling and data-driven decision-making[2].

Such digital twin concepts are increasingly being explored in biomedicine and health care[3–5]. By integrating clinical data, physiological knowledge and computational modelling, biomedical digital twins aim to create dynamic model-driven and data-driven representations of human biology that capture healthy and diseased states. As a systems biology approach, they combine mechanistic models with patient-specific information, allowing simulations of disease progression, mechanistic exploration and prediction of individual health trajectories[6,7]. These capabilities can be translated into precision medicine tools that support quantitative risk assessment, early diagnosis and personalized treatment planning[8–11]. Beyond direct patient care, digital twins could also accelerate drug discovery, reduce drug development costs, and support testing in under-represented or high-risk populations with virtual-patient cohorts[12,13]. As such, biomedical digital twin research has emerged for various physiological systems, including cardiovascular[14,15], neurological[3,16,17], immune[18–20] and oncological domains[9,21–23].

However, the realization of a true medical digital twin remains elusive. Medical digital twins face distinct spatial challenges in representing diseases across molecular to organ scales, as well as temporal challenges in adapting to patient-specific disease progression[24,25]. The progression along the continuum from digital model to digital shadow and digital twin requires advances in model validation, uncertainty quantification[26], algorithmic scalability, continuous data integration and personalization (Fig. 1a). In addition, the model has to provide clinicians with real-time feedback for mechanistic inference, disease forecasting, surgical and procedural planning, and therapeutic optimization (Fig. 1a).

Imaging and monitoring technologies, including echocardiography, computed tomography, magnetic resonance imaging (MRI), wearable biosensors and invasive haemodynamic monitors, can capture structural, functional and physiological data with high spatial and temporal resolution[27], thereby enabling non-invasive and continuous data collection[28]. In addition, algorithmic innovations for multimodel data integration, multiscale physics coupling and artificial intelligence (AI) can provide high-fidelity digital representation and patient personalization. Furthermore, high-performance computing (HPC) approaches and collaborative access to leadership-class supercomputers are expanding the scope of digital twin simulations, allowing near-real-time inference for clinical use[4].

This Review centres on the progression from digital model to digital twin research for transport processes in the human circulatory system, a central driver of physiological homeostasis that supports the function of all organ systems[14] (Fig. 1b). The transport of blood and cells throughout an extensive vascular network makes the system inherently dynamic, complex and multiscale[29] (Fig. 1c). At the macroscopic level, blood flow through large arteries and cardiac chambers governs global haemodynamics; and at the microscopic level, cell–cell and cell–vessel interactions regulate blood cell circulation, metastatic dissemination, and targeted drug delivery within microvascular and porous tissue environments[20,30,31]. These human circulatory transport mechanisms can be computationally modelled to investigate circulatory function[32], disease progression[33,34] and therapeutic response[35,36]. Here, we focus on fluid and cellular transport, which are inherently coupled with other physiological domains, such as cardiac mechanics, electrophysiology and vascular biomechanics. Thus, we also provide context on cardiovascular modelling, highlighting digital twin research that integrates circulatory transport with cardiac function, vascular adaptation and electrical conduction. By situating transport within these complementary areas, we emphasize that advances across interconnected disciplines are essential for the realization of comprehensive and clinically actionable circulatory digital twins.

In this Review, we begin with an overview of circulatory transport pathophysiology on macroscopic and microscopic scales, along with clinical and experimental measurement techniques most relevant to transport-based digital twin construction. We then evaluate circulatory transport digital twin research, assessing both capabilities and limitations. Finally, we discuss the challenges, opportunities and translational considerations that will shape the next generation of digital twins for circulatory transport-related diseases.

Pathophysiology of circulatory transport

The circulatory system is composed of the heart and blood vessels, providing the primary vehicle for the movement of blood, nutrients, cells and oxygen through the body to maintain homeostasis[37–39]. This transport system is also the basis for a spectrum of cardiovascular and haematological diseases.

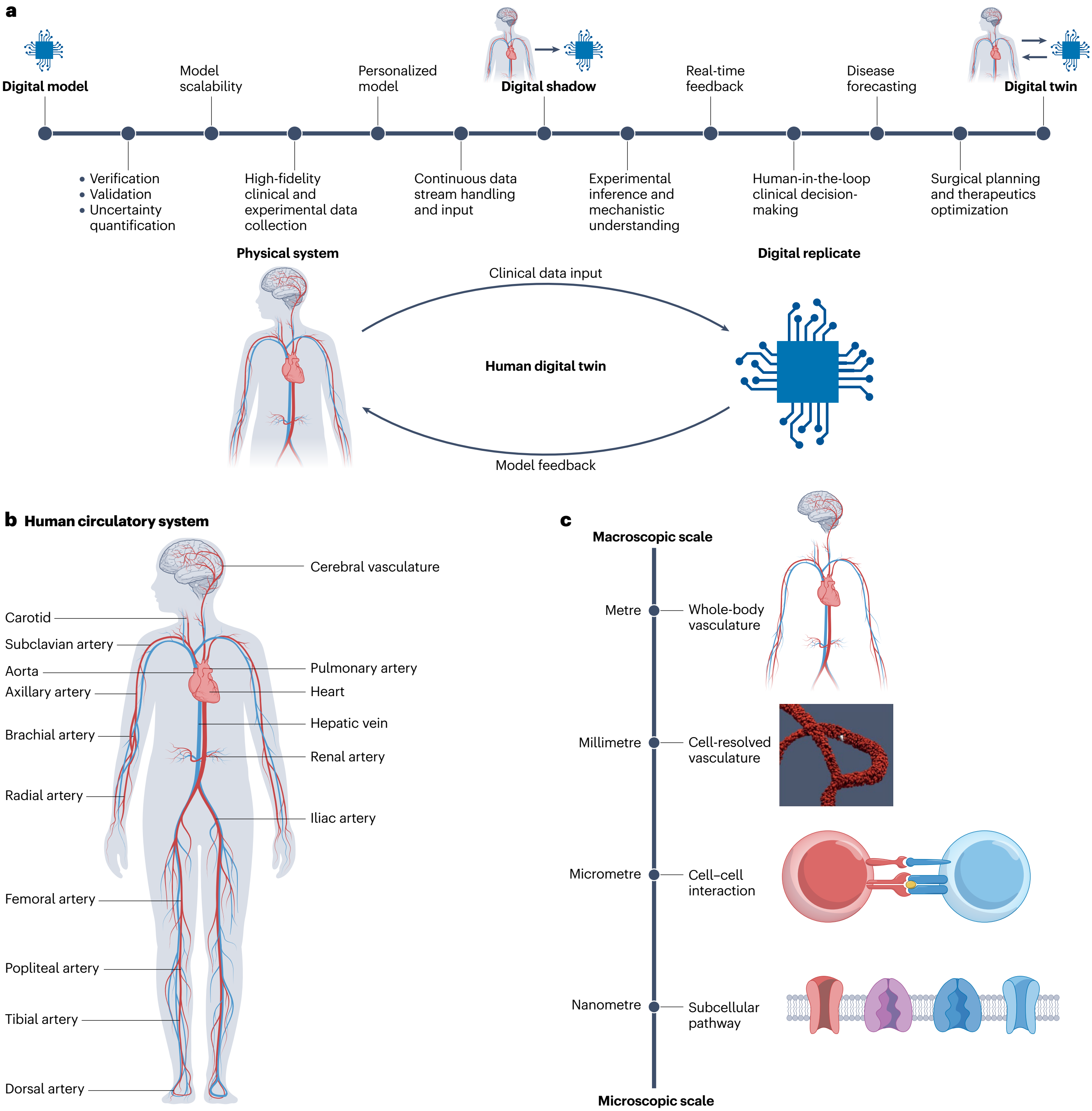
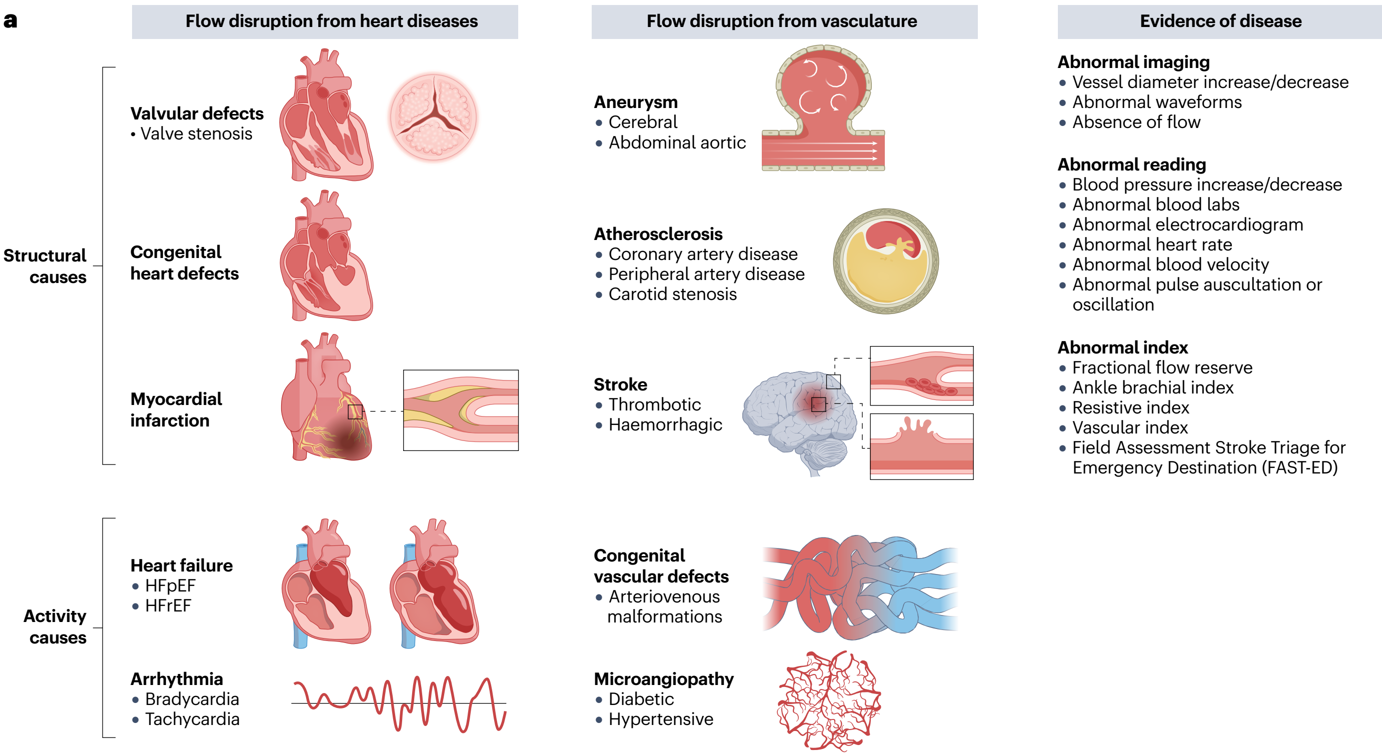
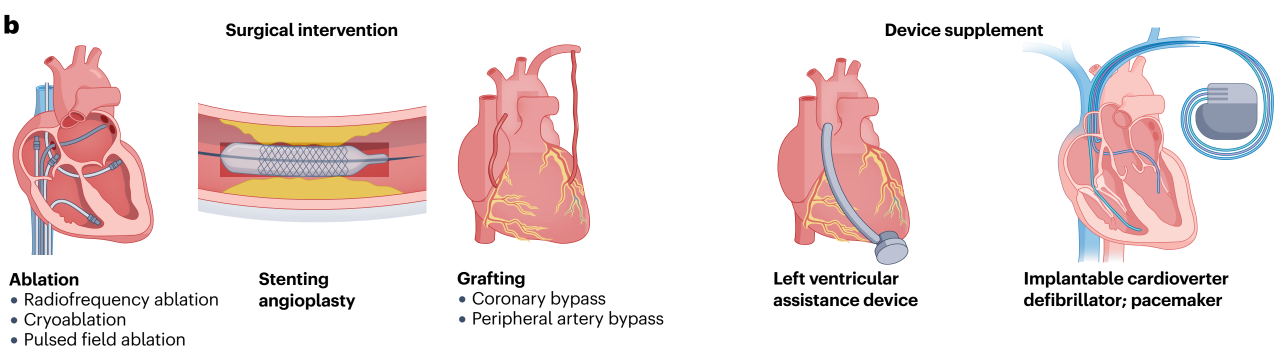


Fig. 1 | Digital twin modelling of the human circulatory system. a, Human digital twins are defined by dynamic two-way coupling between a physical system and its digital replicate. The advancement of digital model to digital shadow and digital twin is a continuum. **b**, The human circulatory system with key vasculatures. **c**, Multiscale physiology of circulation transport, including nanometre-scale subcellular pathways, micrometre-scale cell–cell interactions, millimetre-scale cell-resolved vasculature and metre-scale whole-body vasculature.

Heart diseases related to flow and pump function

Structure-related heart diseases arising from congenital or acquired abnormalities in cardiac anatomy influence circulatory flow[40] (Fig. 2a). For example, valvular diseases can be congenital defects, in which valve stenosis or regurgitation can lead to flow field narrowing and blood backflow. These conditions can also be secondary to other conditions,





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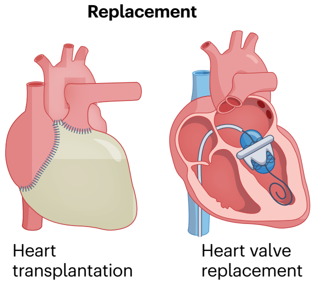


Medication

Antihypertensives	Diuretics, ACE inhibitors, beta-blocker, calcium channel blocker
Anticoagulants	Heparins, vitamin K antagonists, Xa inhibitor, thrombin inhibitor
Anti-arrhythmia	Sodium channel blocker, beta-blocker, potassium channel blocker, calcium channel blocker
Lipid-lowering medications	Statins, bile acid sequestrant, niacin, fibrates, PCSK9 inhibitor

Lifestyle

- Diet
- Smoking cessation
- Stress reduction
- Exercise

Fig. 2 | Macroscopic overview of human circulatory transport. a, Flow disruption due to disease states originating from the heart and vasculature. Heart disease can be classified by structural causes and activity causes. Structural causes include valvular defects, congenital heart defects and myocardial infarction. Activity causes include heart failure and arrhythmia. Vascular diseases include aneurysm, atherosclerosis, stroke, congenital defects and microangiopathy. Cardiovascular disease can be diagnosed by imaging and the measurement of key parameters. **b**, The clinical treatment for circulatory diseases includes surgical intervention, device supplement, replacement, medication and lifestyle changes. ACE, angiotensin-converting enzyme; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; PCSK9, proprotein convertase subtilisin/kexin type 9.

such as ageing, hypertension, myocardial infarction, cardiomyopathies or postoperative repair[41,42]. Other congenital heart defects, such as tetralogy of Fallot and hypoplastic left heart syndrome, are characterized by malformations of the heart chambers or great vessels, altering the direction of blood flow[43,44]. Digital twins can guide patient-specific therapy for these conditions by reconstructing an individual's cardiac anatomy from imaging data, supporting precise surgical planning.

Myocardial infarction is a frequent cause of pathologic structural remodelling of the heart; here, ischaemia caused by blocked coronary blood flow causes irreversible cardiomyocyte death, leading to ventricular scar formation[45,46]. In addition, post-myocardial infarction fibrosis can result in electrically inert, non-contractile tissue that can alter heart function and progress to heart failure[47]. Therefore, prompt diagnosis and treatment are vital. Digital twin tools can aid in the stratification of infarct risk and model post-myocardial infarction complications, such as valve dysfunction and aberrant ventricular remodelling, which can impact systemic flow conditions[14].

Beyond structural abnormalities, functional disturbances in cardiac activity, such as heart failure and electrical disorders, known as arrhythmia, can affect myocardial contractility and alter haemodynamic flow patterns, including reduced cardiac output. Heart failure is defined by inadequate blood supply to meet the metabolic demands of the body, and is often caused by left-ventricular dysfunction owing to infarction-related fibrosis, ischaemic heart disease, chronic hypertension or valvular heart disease[48]. In advanced cases, left-ventricular assist devices may be required to support myocardial contraction until heart transplantation[49] (Fig. 2b). Heart failure is difficult to monitor and diagnose owing to high variations in clinical aetiology and disease progression. Digital twin approaches could aid in clinical diagnosis to classify heart failure phenotypes and guide patient-specific treatment[14].

Arrhythmias arise from abnormalities in the heart's electrical conduction system that disrupt coordinated chamber contraction[50]. Such electrical dysfunction can diminish pump efficiency and alter haemodynamic flow patterns. As arrhythmias are heterogeneous and often intermittent, their haemodynamic impact is challenging to assess[50]. Digital twin-based risk stratification offers a means to evaluate downstream flow disturbances and guide management[14].

Disrupted blood vessels

Vascular integrity is as important as cardiac function; vessels supply blood, oxygen and nutrients to the entire body. An integral component of this homeostatic maintenance is vascular autoregulation, that is, the intrinsic ability of blood vessels to adjust their diameter to maintain consistent blood flow despite changes in perfusion pressure. This regulation is mediated by coordinated mechanisms, including myogenic response of vascular smooth muscles to stretch, adrenergic control through sympathetic innervation and metabolic signalling that corresponds to tissue demand. The baroreflex complements these local mechanisms by adjusting heart rate and vascular tone in response to systemic changes in blood pressure[51,52]. These processes influence vascular remodelling, that is, the structural reorganization of blood vessel walls through processes such as smooth muscle cell migration, extracellular matrix turnover and endothelial adaptation, ensuring the long-term adaptation of vessel diameter and function. In pathological conditions, these processes can be altered in compensatory efforts or secondary to distorted signalling, leading to disrupted or deleterious remodelling[53,54].

Disruption of this homeostasis can lead to cardiovascular events, such as stroke[55,56] (Fig. 2a). Stroke can arise from vessel rupture or vascular obstruction, which is known as haemorrhagic or ischaemic stroke, respectively. Haemorrhagic stroke results from bleeding in the brain, often caused by rupture of a cerebral aneurysm, localized dilation of a weakened cerebral artery wall or loss of structural integrity of the microvasculature[57,58]. By contrast, ischaemic stroke occurs when blood flow to the brain is blocked, most often by an obstructive clot[58,59]. As the cardiovascular system works in homeostatic balance, disruptions in one region can increase the risk of other states of disease. For example, the likelihood of both types of stroke increases in the presence of microangiopathy from diabetes or hypertension, as well as other vascular abnormalities, such as arteriovenous malformations and congenital vascular disorders[60]. Similar to stroke, pathological vessel dilation can also lead to a thoracic or abdominal aortic aneurysm, conditions that carry a substantial risk of rupture, leading to internal bleeding and poor clinical outcomes[61].

Atherosclerosis, which is marked by the build-up of lipids, inflammatory cells and calcium-rich plaques that narrow the lumens of vessels, such as the carotid, coronary and peripheral arteries[62–64], is a substantial contributor to cardiovascular events. For example, carotid artery atherosclerosis is a major cause of ischaemic stroke. In the coronary arteries, plaque-induced luminal narrowing or occlusion reduces myocardial oxygen delivery by restricting blood flow[64]. Peripheral artery disease shares the same plaque pathology and frequently coexists with coronary artery disease as both disease states share similar risk factors, such as hypertension, diabetes, tobacco use and hypercholesterolaemia[63]. These vascular morbidities can be treated by interventions such as angioplasty, stenting and bypass grafting surgery (Fig. 2b), which can be guided and optimized by digital twins in a personalized manner[36,65,66].

Circulating blood cell diseases

On the microscopic scale, the components of the blood are central to circulatory physiology and disease (Fig. 3a). Blood consists of plasma and circulatory cells, including red blood cells, white blood cells and platelets, which collectively support oxygen transport, haemostasis, and inflammatory and immune defence. Dysfunction in any of these components can contribute to a range of circulatory disorders and affect organ function.

Red blood cells are the most abundant blood cells, transporting oxygen through haemoglobin. These cells are adapted for efficient gas exchange through their small size, flexible membrane and large surface area. The loss of red blood cells or poor red blood cell formation results in anaemia, a reduction in oxygen-carrying capacity that causes tissue hypoxia. Sickle cell anaemia, an inherited disorder caused by abnormal haemoglobin, is characterized by rigid, elongated red blood cells that undergo haemolysis and obstruct vessels, leading to inflammation, vascular injury and tissue ischaemia[67]. Furthermore, changes in the properties of ageing red blood cells can contribute to a decline in health, such as a decrease in the integrity of the blood–brain barrier[68]. Platelets, another essential blood component, maintain vascular integrity through haemostasis. When vessel injury occurs, platelets adhere to the damaged site, become activated, and aggregate to form a platelet plug while promoting fibrin deposition and repair[69]. Dysfunctional platelets can cause bleeding or pathological thrombosis, with potential complications such as ischaemia or stroke. Beyond clotting, platelets also participate in inflammatory processes[70], reflecting their influence on circulatory pathology.

White blood cells, or leukocytes, are key to immune surveillance and host defence[71]. Neutrophils are the most abundant white blood cells, responding rapidly to bacterial infection through phagocytosis. Monocytes, recruited to sites of tissue damage by cytokines and chemokines, adhere to vessel walls, migrate into affected tissues, and subsequently differentiate into macrophages that remove debris and present antigens. B and T lymphocytes then recognize antigens, produce antibodies and destroy infected cells. The disruption of leukocyte function can lead to excessive inflammation or increased susceptibility to infection.

Blood cell counts and properties measured in routine blood tests are often used to diagnose and monitor disease. Digital twins could provide a mechanistic understanding of the interactions between blood cell components and improve risk stratification of patient-specific blood disorders.

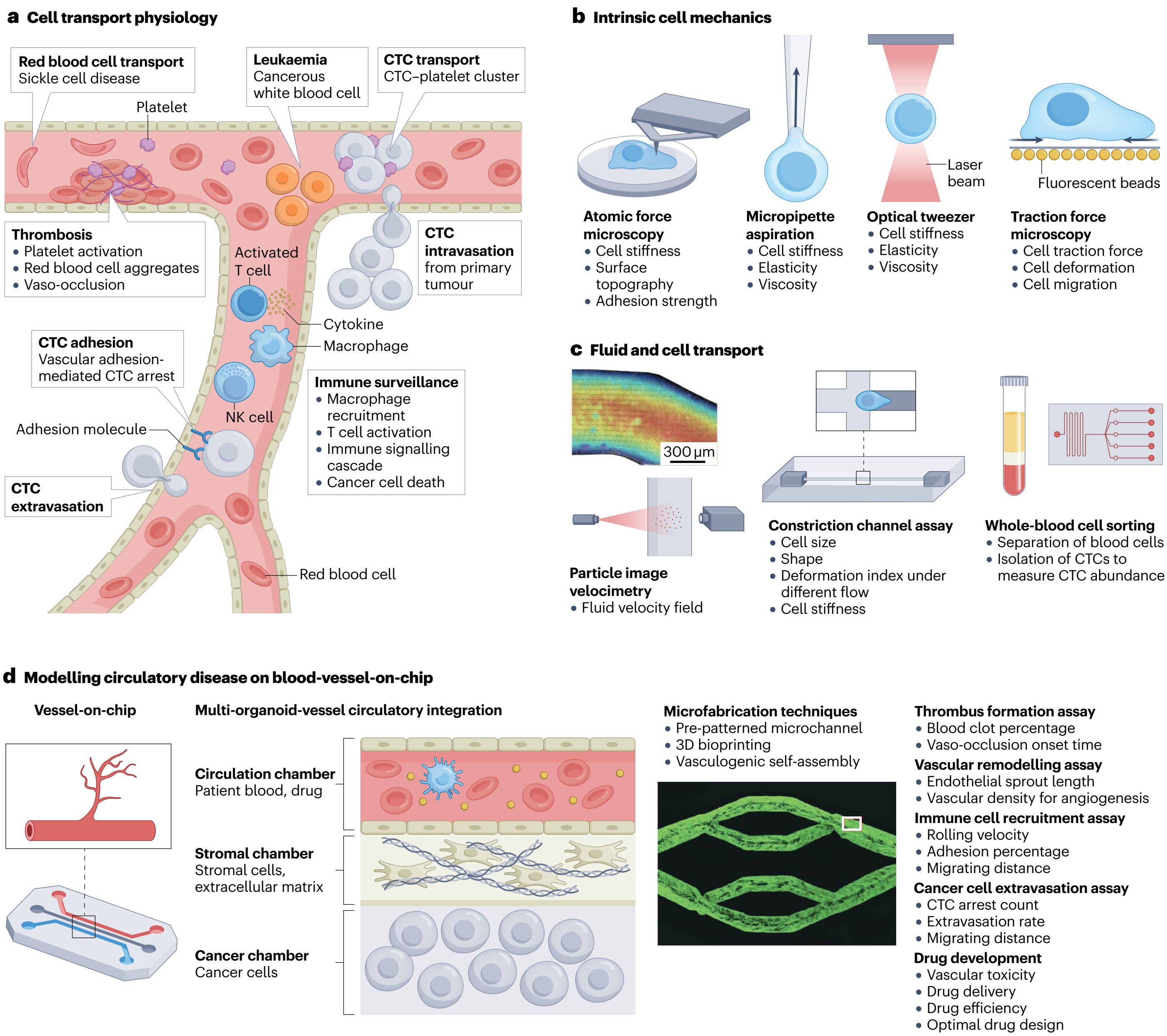


Fig. 3 | Microscopic overview of human circulatory transport. a, Transport and cellular interactions of key circulatory cells, including red blood cells, platelets, white blood cells and circulating tumour cells (CTCs). **b**, In vitro assessment of cell-intrinsic material properties. **c**, Fluid and cell mechanics measurements in microfluidic systems. Particle image velocimetry characterizes the flow field[194]. Cell deformation under flow conditions can be assessed using imaging-based approaches in constriction channel assays. Microfluidic devices can be applied to sort and characterize cell properties across different cell types. **d**, Vessel-on-chip systems can mimic cell transport in the vasculature[194]. Multi-organoid-vessel circulatory integration can be achieved by culturing different cell types in designated chambers using microfabrication techniques. Vessel-on-chip tools can be applied for thrombus formation assays, vascular remodelling assays, immune cell recruitment assays, cancer cell extravasation assays and drug development. NK, natural killer. Panels **c** (particle image velocimetry) and **d** (microfabrication techniques) reprinted with permission from ref. 194, AAAS.

Table 1 | Clinical measurement and inference for digital twin modelling

Metric	Clinical measurement (non-invasive)	Clinical measurement (invasive)	Incorporation into digital twins
Pressure	Cuff measurement Wearable devices	Cardiac catheterization	Inform boundary conditions, validate CFD studies as a tuning factor, predict disease diagnosis and progression
Heart electrophysiology	Electrocardiogram Holter monitor Photoplethysmography	Intracardiac electrophysiology	Reconstruct patient-specific electrophysiological maps, inform arrhythmia risk stratification
Blood flow	Catheterization Four-dimensional magnetic resonance imaging Doppler ultrasound Echocardiography	–	Infer pulsatile input waveform, inform inlet and outlet boundary conditions, provide validation for CFD simulations
Vessel and heart structure	Angiogram Computed tomography angiography Magnetic resonance Angiography/imaging X-ray Echocardiography	Cardiac catheterization Optical coherence tomography	Reconstruct patient-specific anatomy from image segmentation, inform vascular geometry in CFD simulations, assess patient risk and aid surgical planning
Cell properties	Atomic force microscopy Micropipette aspiration Optical tweezers Traction force microscopy Microfluidic constriction channels	–	Infer cell size and shape in model, infer cell viscoelastic properties (stiffness, elasticity and viscosity) for cell mechanics modelling, validate cell models
Cell transport in flow	Microfluidic platforms Vessel-on-chip	–	Reconstruct realistic vasculature, inform cell trajectory

CFD, computational fluid dynamics.

Cancer metastasis

The transport of circulating tumour cells plays a pivotal role in cancer metastasis. Circulating tumour cells are rare in blood, with around one to ten circulating tumour cells per millilitre of blood in patients with metastasis. However, they are detectable in many types of cancer and might provide a biomarker and prognostic tool[72]. Circulating tumour cells originate from a primary tumour, enter the bloodstream and travel to distant organs where they can establish secondary tumours (Fig. 3a). Although most circulating tumour cells are eliminated by shear stress or anoikis, some survive by forming clusters with neutrophils, platelets or cancer-associated fibroblasts, thus enhancing immune evasion and metastatic potential[73]. Distinct circulating tumour cell phenotypes can influence circulation patterns and preferential colonization of specific distant organs[74]. The arrest of circulating tumour cells on the vessel wall and their extravasation from the lumen of blood vessels are shaped by the surrounding microenvironment, including blood flow haemodynamics, adhesive interactions with the vasculature and margination effects from red blood cells[75]. These mechanisms, and the related metastatic potential, can be assessed by digital models[76,77].

Multiscale clinical and experimental inputs

Clinical and experimental measurements are essential for building accurate and physiologically relevant digital twins of the circulatory system. Quantitative measurements can be used to parameterize, personalize and validate digital twins that simulate both healthy and disease conditions on macroscopic and microscopic scales. These measurements capture key physiological and structural attributes, including blood pressure, flow characteristics, cardiac and vascular anatomy, and cellular morphology, which together form the crucial parameters and model assumptions for personalized digital twins of circulatory transport (Table 1).

Haemodynamic parameter collection

Blood pressure is a central haemodynamic indicator used in cardiovascular digital twins to define boundary conditions or as a tuning target in computational fluid dynamics simulations[78] (Fig. 2a). Non-invasive methods, such as oscillometric or auscultatory blood pressure cuff measurements of the brachial artery, provide a surrogate measure for systemic pressure. However, certain pathologies require invasive blood pressure assessments that are obtained at the site of vascular disease. For patients with coronary artery disease, fractional flow reserve is the gold standard diagnostic metric to determine whether coronary artery stenosis is flow limiting and likely to cause ischaemia. Fractional flow reserve is measured at the time of invasive coronary angiography and is derived from pressure readings taken on both sides of a stenotic lesion[79]. Localized measurements are based on a pressure wire, whereas intracardiac pressures, such as those monitored in patients with heart failure, are measured using a catheter that is introduced through a peripheral vein and advanced to the cardiac chambers, where pressures are recorded[80,81].

In addition, electrophysiological measurements can give insight into the pumping efficacy and mechanism of the heart. These readings are essential for informing and personalizing digital twin models that involve fluid and heart activity, enabling accurate simulation of electrical conduction influencing heart contraction performance and pulsatile blood flow dynamics. Surface and wearable sensors that record electrocardiogram (ECG) and photoplethysmography (PPG) signals can provide temporal markers of cardiac contraction and relaxation that define the rhythm or amplitude of pulsatile flow[82,83] (Fig. 2a). Standard 12-lead ECGs and long-term Holter or patch recordings can be used to characterize heart rate dynamics and estimate variations in stroke volume and cardiac output over time[84]. PPG offers continuous assessment of flow pulsatility and vascular compliance, serving as a key input for modelling systemic flow resistance and perfusion. Wearable and implantable sensors further extend these measurements by capturing real-time data on blood pressure and cardiac output, providing high-resolution temporal patterns of flow generation[82,85]. Together, these measurements provide crucial inputs for digital twin frameworks that seek to reproduce and predict patient-specific blood transport and pulsatile flow dynamics.

Circulatory system anatomical reconstruction

Structural and functional information of the heart and vessels are important for patient personalization of 3D digital twin models. Computed tomography provides cross-sectional views of organs, vessels and soft tissue, often enhanced with contrast to detail the vasculature[86] (Fig. 2a). Angiography is used to visualize arterial and venous structures, typically by injecting a contrast agent into vessels that can be visualized by X-rays[87]. Intravascular ultrasound and optical coherence tomography provide high-resolution intravascular imaging for detailed evaluation of the vessel lumen and intravascular plaque characteristics[88]. Images from these modalities allow patient-specific mesh generation and volumetric reconstruction of vascular networks from planar segments, which can serve as a geometric foundation for 3D digital twin frameworks.

Imaging modalities, such as MRI, can also capture functional and dynamic information[89], for example, to quantify myocardial strain, heart wall motion and contractility[90]. Furthermore, time-resolved four-dimensional MRI can provide velocity fields and flow vectors to inform boundary conditions of digital twin simulations[91]. Additional flow-related data can be acquired using Doppler ultrasound, a non-invasive modality that uses sound waves to assess blood flow through vessels[92]. Echocardiography, that is, ultrasound applied to the heart, can evaluate valve function, identify regurgitation, stenosis and congenital abnormalities, and assess the overall size and morphology of the heart[93].

Cell property characterization

High-fidelity cell mechanics characterization can capture cellular function and disease progression[94]. In static cell culture, cell mechanics can be assessed by atomic force microscopy, micropipette aspiration, optical tweezers and traction force microscopy (Fig. 3b). In atomic force microscopy, depth-sensing indentation is applied to mechanically probe adherent cells, providing high-resolution measurements of stiffness (for example, Young's modulus), surface topography and adhesion strength[95]. Micropipette aspiration applies suction to suspended cells, measuring membrane deformation to assess elastic and viscous properties, complementing the ability of atomic force microscopy to analyse the whole cell or localized mechanics[96]. The optical tweezer uses a highly focused laser beam to investigate local viscoelasticity and molecular bonds, whereas optical stretchers, which are related to optical tweezers, can deform entire nucleated cells[96]. Traction force microscopy quantifies cell-generated forces by measuring substrate deformations, enabling the study of processes, such as migration. This method is independent of cell size and force magnitude, thereby supporting a range of applications[97].

Cell mechanics in flow

Microfluidic platforms allow real-time analysis of circulatory cells under flow (Fig. 3c). Integrated particle image velocimetry provides velocity and shear profiles to understand the flow field in microfluidic devices[98]. Mechanical properties can be assessed by measuring cell deformation and transit time through constriction channels under flow through hydrodynamic or shear-based deformation[96]. These constriction channels can be actuated by electromagnetic, optical or acoustic forces. These assays can also function passively driven by cell adhesion, filtration or intrinsic physical characteristics, such as cell size, shape or density[96,99]. These systems support high-throughput, label-free characterization of heterogeneous samples and circulating tumour cell movement in vascular systems[100,101]. Another approach in microfluidic applications is image-based cell sorting, which can be applied to classify and sort cells by analysing the morphological, organelle and interaction features of cellular images in dynamic microenvironments[102]. Microfluidics is further clinically applied to sort blood cells by stiffness, size or dielectrophoretic response by measuring transit time, light intensity and flow velocity to derive metrics such as deformability, aggregation and viscosity[103].

Cell dynamics monitoring

Measuring cell interactions in a realistic model of the vasculature is essential to validate and inform digital twins. For example, vessel-on-chip systems combine microfluidic devices made from materials that mimic blood vessel properties with vascular cells, providing a key platform to investigate cell behaviour under physiological flow conditions in a biomimetic environment[104] (Fig. 3d). Furthermore, patterned microchannel fabrication and 3D bioprinting enable in vitro vascular models that replicate complex anatomies, such as aneurysms, stenoses and organ-scale networks[105,106]. Perfusable vessel-on-chip systems exploit vasculogenic self-assembly and hydrogel-based microchannels to support functional endothelial cell cultures[107]. Their physiologically relevant geometry and flow conditions allow the study of a range of circulation-related phenomena (Fig. 3d), including vascular remodelling and disease progression, thrombosis and immune cell recruitment[104]. These systems can also be integrated with organ-on-a-chip systems to create vascularized tissue and multi-organ models. For example, vascularized tumour cultures have been used to study tumour angiogenesis, cancer metastasis, drug screening and drug responses, using cancer cells from patients to facilitate the development of personalized therapies[104].

Computational modelling for digital twins

Modelling approaches in digital twin frameworks can be broadly categorized into model-centric and data-centric approaches. Model-centric digital twins rely on mechanistic, physics-based representations of physiology, offering high fidelity and well-defined parameter spaces. However, constructing multiscale, physics-based models typically comes with substantial computational demands as the spatial and temporal domains grow. By contrast, data-centric twins leverage statistical learning and AI-driven surrogates to emulate system behaviour directly from observational data. They enable rapid inference and require fewer computing resources during real-time deployment; however, they face barriers in verification, validation and uncertainty quantification, particularly when extrapolating beyond observed data.

Advancements in computing techniques and resources are essential for enabling both approaches. Leadership-class HPC systems, heterogeneous central processing unit (CPU)–graphics processing unit (GPU) architectures, domain decomposition, load balancing, mixed-precision GPU kernels, and overlapped input and output can extend the spatial and temporal resolution of model-centric simulations, while accelerating training and deployment of data-centric surrogates.

Model-centric approach

Model-centric digital twins for circulatory transport are grounded in physics-based representations of fluid dynamics and cell–vessel interactions. These frameworks typically couple the Navier–Stokes equations with constitutive models for blood rheology, vessel wall mechanics and particulate transport, enabling high-resolution prediction of haemodynamics and cellular trajectories (Fig. 4a). For large-vessel haemodynamics, finite-element and finite-volume Navier–Stokes solvers such as SimVascular and CRIMSON support patient-specific boundary conditions and fluid–structure interactions. OpenFOAM provides open-source flexibility for custom rheology and multiphysics extensions, whereas commercial platforms such as ANSYS Fluent are commonly used for device evaluation, surgical planning and valve mechanics[108]. At the microscale, lattice Boltzmann method-based solvers including HemeLB, HemoCell and HARVEY[109] resolve dense suspensions of deformable cells, enabling high-performance simulations of microvascular networks and disease-specific transport of cells such as circulating tumour cells. Particle-based frameworks such as LAMMPS and Palabos-npFEM[110] extend modelling fidelity to detailed red blood cell mechanics and large-scale platelet transport.

Together, these solvers form a versatile toolkit for bridging scales, from organ-level circulation to subcellular biomechanics, and provide the mechanistic foundation upon which hybrid and AI-accelerated digital twin approaches are being built.

Data-centric approach

Data-centric methods learn statistical relationships between anatomy, physiology and haemodynamics, providing fast and scalable approximations in situations in which physics-based simulations are computationally demanding (Fig. 4b). These approaches are important for circulatory-transport digital twins, which must integrate multimodal data, adapt to evolving physiology and operate at clinically relevant timescales to inform diagnosis, guide treatment decisions and update patient-specific predictions within the time frame of clinical workflows. Such workflows often require time frames from minutes (for example, intra-procedural guidance and emergency unit monitoring) to hours or days (for example, treatment planning and risk stratification). Therefore, digital twins must generate meaningful outputs within these actionable windows.

A first class of data-centric methods relies on supervised surrogate models that learn pointwise relationships between anatomy, inputs and haemodynamic outputs. These data-driven surrogates, such as feedforward models, convolutional neural networks and graph neural networks, are trained on large libraries of computational fluid dynamics simulations to rapidly predict flow, pressure and wall shear stress[111–113], allowing non-invasive fractional flow reserve estimation, coronary flow prediction and assessment of shear-related plaque risk. However, they are limited by training data diversity and lack of reliability. In addition, time-series models analyse physiological waveforms that influence circulatory transport. For example, recurrent neural networks, long short-term memory networks, temporal convolutional networks and transformer architectures can be applied to ECG, PPG and blood pressure signals to estimate arrhythmia burden, autonomic changes and dynamic boundary conditions, such as cardiac output[114]. These models support digital twins by providing real-time physiological drivers; however, they are sensitive to distribution shift and can struggle with causal interpretation in complex pathophysiology.

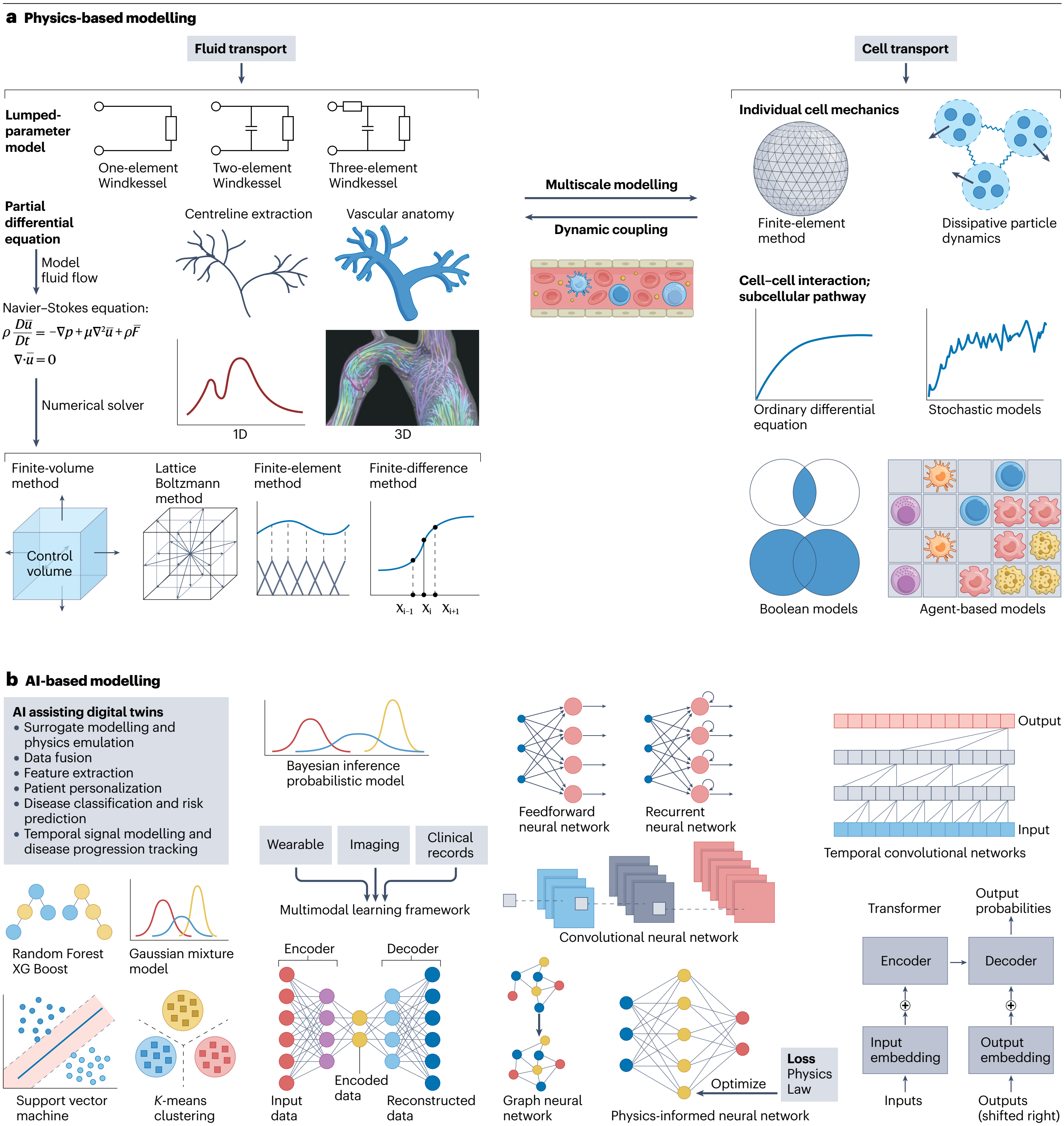
A second, more general class of models focuses on an operator learning the full functional mapping, from geometry and boundary conditions to spatiotemporal flow fields. Unlike pointwise supervised surrogates, operator-learning approaches, such as Fourier neural operators and DeepONet, can learn mappings from functional inputs, including geometry or boundary waveforms and full spatiotemporal haemodynamic fields[115,116]. This allows rapid approximations of solutions to Navier–Stokes-based models and generalization across discretizations. However, operator learners remain sensitive to training range and require careful validation to ensure that subtle but clinically meaningful flow features (for example, low-shear pockets and oscillatory shear) are preserved. Physics-informed models, including physics-informed neural networks and physics-regularized neural operators, impose governing equations as soft constraints[117–119]. These approaches are promising when data are sparse but physiology is well understood, enabling the reconstruction of pressure or velocity from limited measurements. However, their optimization and scaling to full 3D cardiovascular domains remain difficult and their convergence is slow.

Finally, foundation models for fluid dynamics rely on large pretrained neural operators or transformer-based physics models that learn generalizable flow priors across diverse geometries (for example, Poseidon, PROSE-FD and other partial-differential-equation-based foundation models)[120,121]. Developed largely within broader computational fluid dynamics (CFD) and multiphysics modelling communities, these approaches demonstrate how large-scale pretraining can accelerate downstream flow prediction and support cross-task generalization. This approach is being explored for cardiovascular modelling, with operator-learning frameworks and equivariant graph-based haemodynamic estimators showing promise but remaining limited in their ability to transfer across anatomies[122]. Importantly, true foundation-scale cardiovascular models have not yet been developed owing to the scarcity of large, diverse circulatory flow datasets and the need for stringent guarantees of physical fidelity, numerical stability and clinical uncertainty quantification. As these resources mature, foundation-model paradigms may become a key accelerant for circulatory digital twins.

High-performance computing

Advances in computing hardware have been central to expanding the modelling scope for model-centric and data-centric frameworks. Modelling initially relied on massively parallel distributed memory systems, in which large problems are decomposed into thousands of compute nodes. Parallelization frameworks, such as the message passing interface, enable efficient communication, making metre-scale vascular domains and multiscale physiological models computationally tractable[123,124]. By contrast, GPUs allow fine-grained parallelism. Originally designed for computer graphics rendering, GPUs have become general-purpose accelerators optimized for high-throughput, data-intensive tasks. Programming models, such as CUDA, SYCL and OpenCL, rely on CPUs to orchestrate control logic, while offloading numerically intensive kernels to GPUs. Leadership-class systems, such a El Capitan, Frontier, Aurora and JUPITER Booster, combine CPUs, GPUs and specialized accelerators into hybrid architectures capable of exascale performance, allowing faithful digital twin simulations[125].

Exploiting this hardware requires deep integration of models with architecture-aware algorithms through domain decomposition,



memory-aware and cache-aware data layouts, mixed-precision kernels and performance-portable code. Coupled with high-performance input, output and load balancing, these strategies allow near-real-time simulations, for example, for whole-body blood flow simulations of the arterial system[126] and large-scale red blood cell models that encompass millions to trillions of cells[127–129]. Several of these efforts earned recognition in the form of the ACM Gordon Bell Prize, establishing benchmarks for biomedical computing and forming the technical foundation for building digital twins.

Human circulatory digital twin applications

Achieving a fully realized circulatory digital twin remains an aspirational goal. However, digital models can offer mechanistic insights and accurately represent circulatory dynamics. In addition, digital shadows

Fig. 4 | Computational modelling of the human circulatory system. **a**, Physics-based modelling for fluid mechanics and cell mechanics. Fluid transport modelling typically uses a reduced-order lumped-parameter model or 3D simulations solving the Navier–Stokes equation. Common solvers for the Navier–Stokes equation include the finite-volume method, lattice Boltzmann method[195], finite-element method and finite-difference method. Individual cell mechanics can be modelled by the finite-element method and dissipative particle dynamics. In addition, cell–cell interactions can be modelled using ordinary differential equation, stochastic models, Boolean models and agent-based models, depending on the specific application. Fluid and cell mechanics can further be coupled by fluid–structure interaction modelling using the immersed boundary method[195]. **b**, Key application objectives for artificial intelligence (AI)-assisted digital twins. Surrogate modelling and representative algorithm families can be used to support these objectives, including Bayesian inference, machine learning, deep neural networks (convolutional, recurrent, temporal convolutional and transformer based), graph neural networks, autoencoders and physics-informed neural networks. Panel **a** (lattice Boltzmann method) reprinted from ref. 195, Springer Nature Limited.

and digital twin frameworks can aid in clinical decision-making, for example, for the prediction of disease outcomes, risk assessment, long-term prognosis, surgical planning and personalized treatment optimization (Table 2).

Mechanistic insights

Understanding mechanisms and pathways in biological systems provides the building block of high-fidelity digital twins. Computational modelling has great merit in studying the complex, multiscale circulatory system in a controlled manner, isolating the effect of different pathways to reconstruct correlation and causation from biomechanical and biomolecular perspectives. For example, computational modelling has been applied to probe vascular haemodynamics, revealing that altered stress fields are directly associated with specific disease conditions[130–133]. Importantly, computational simulations of disease mechanisms can uncover hidden phenotypes in disease conditions, for which clinical measurements are inaccessible for dynamic monitoring.

CFD simulations have shown that various cardiovascular diseases have altered flow rates, velocity fields and shear stress distributions[133–136]. For example, digital models of complex vascular lesions have identified variations in high wall shear stress[137], suggesting new prognostic metrics beyond conventional pressure-based indices[131,138]. In addition, computational investigations of vessels prone to aneurysms have demonstrated that increased viscosity, in conjunction with elevated wall shear stresses, may contribute to the risk of vessel rupture[139]. Comparable mechanistic findings have been reported for plaque stability, indicating an association between high wall shear stress, oscillatory shear index and carotid plaque rupture[32]. Furthermore, simulations captured haemodynamic alterations in the left atrial appendage in patients with atrial fibrillation, revealing that decreased flow speed and increased flow oscillation contribute to an increased risk of thrombus formation[34]. These digital models show that haemodynamic features, such as wall shear stress, which are difficult to estimate in vivo, can be systemically studied to uncover disease mechanisms and predictive biomarkers.

At the microscopic level, digital twins enable the investigation of biomechanical mechanisms underlying red blood cell deformation[140], haemolysis[141] and various red blood cell diseases[30,142]. For example, the immersed boundary method has been applied to model alterations in red blood cell transport in sickle cell disease, COVID-19, sepsis, iron deficiency anaemia and spherocytosis[142]. Here, red blood cells are modelled as fluid-filled deformable capsules, inferring that their high stiffness causes fluctuations in wall shear stress in these blood disorders[142]. Haemodynamic alterations owing to stiff red blood cells have also been predicted in an endothelialized microfluidic device, connecting sickle cell disease with endothelial inflammation and dysfunction[30]. Furthermore, platelet adhesive dynamics and coagulation processes in diabetic blood have been investigated within idealized channel models[143]. This digital model was also applied to clinically observed retinal microaneurysms, demonstrating that thrombosis is more likely to initiate in conditions of prolonged platelet residence time[144].

Computational modelling of circulating tumour cell transport has advanced our understanding of metastatic site preferences[145,146]. These computational models for cancer transport have proven to accurately capture cell deformation through microcirculatory flow, allowing the analysis of the mechanical behaviour of different cell types during metastasis[147]. Flow-perfused microfluidic systems enable the investigation of how distinct adhesive molecules regulate circulating tumour cell arrest, supported by trajectory analyses of mouse carcinoma cells under various ligand depletion conditions[76]. Circulating tumour cell adhesion has also been studied in vasculature with stenosis and with varying vascular glycocalyx compositions on a vessel-on-chip platform[77]. In addition, CFD can be combined with endothelialized microvessels to assess how local haemodynamics and cell deformability regulate attachment at bifurcations and influence residence time[148]. The transport of red blood cells and circulating tumour cell clusters has been modelled using imaging data from patients with breast cancer, suggesting that circulating tumour cell aggregates with reduced membrane elasticity enhance rolling and vascular interactions[149]. Therefore, integrating CFD with microfluidic and digital twin platforms can aid in the investigation of the biomechanical determinants of metastatic cell transport. Although these digital models typically lack patient personalization and continuous data integration, they can provide mechanistic insights into the circulatory system and lay the foundation for advancing to a high-fidelity virtual representation in the continuum from digital model to digital twin.

Circulatory dynamics

To accurately capture human circulatory transport, digital twins must integrate mechanistic knowledge, informed by the clinical application. For example, although vascular blockage may be captured by fluid dynamics alone, modelling metastasis requires the inclusion of cell transport and interactions. Thus, digital twins should balance biological complexity with computational efficiency, incorporating only the processes essential for the target application.

Microfluidic devices benefit from controlled and well-defined fluid fields, which can be modelled by digital twins for device optimization. For example, lumped-parameter modelling has been combined with finite-element method to design micropump-driven flows in microfluidic systems[150]. Similarly, CFD has been applied to translate vessel-level haemodynamics into a vessel-on-chip platform to study cerebral sinus thrombosis[151]. However, moving from engineered platforms to patient-specific vasculature, model-centric digital twins face challenges in integrating relevant physiology and accommodating dynamically changing systems. Nevertheless, static digital models and dynamic digital shadows can accurately reproduce blood flow

Table 2 | Digital twins of human circulatory transport

Model inference	Clinical and experimental input	Model algorithm	Requirements to qualify as digital twin	Refs.
Mechanism of diseased red blood cell transport and endothelial dysfunction	Cell stiffness, diseased red blood cell fraction	Projection method, IBM	Integration of patient-specific red blood cell profile for risk stratification	30,142
Red blood cell and platelet transport mechanism for coagulation in diabetic blood	Patient data informs cell stiffness, as well as red blood cell and platelet composition in healthy and diabetic blood	DPD, Morse potential for cell–cell adhesion	Long-term blood monitoring to inform coagulation location in patient; transport prediction in large, realistic vasculature	143,144
Mechanistic insight into circulating tumour cell adhesive transport	Microfluidics of mouse breast carcinoma cells with seven depletion patterns of surface adhesive ligands	Oscillating Poiseuille model and an ODE-based adhesion model	Limited resolution for cell trajectory recording; challenges in model validation	76
Prediction of aortic blood flow	Four-dimensional MRI-informed patient vascular geometries (n=40) and synthetic geometries (n=1,000)	Vascular modelling toolkit, deformetrica, CFD, DNN	Longitudinal domain validation to capture temporal haemodynamic changes	160
Multiphysics digital model with coupled cardiac electromechanics and vascular haemodynamics	Imaging-reconstructed left-ventricular mesh and thoracic aorta, haemodynamics-informed boundary conditions and flow conditions	Electromechanical model of heart in Alya, fluid mechanics model of blood flow in HemeLB, file-based coupling scheme between Alya and HemeLB	Additional assessment for the ability to achieve real-time simulations and predictions; continuous integration of cardiac and haemodynamic measurements	163
Digital heart with modelled muscle contraction, pathological conditions and cardiac resynchronization therapy	Patient-specific heart structure and valve position, activation pathway, blood pressure	Fung-type model, IBM, fluid structure electrophysiology interaction solver	Improve computational costs to capture temporal changes of heart activity	164
Mechanistic insight into adhesive circulating tumour cell and red blood cell transport	Cell material properties	IBM, FEM, LBM, APR	Integration of patient-specific cell profile for risk stratification	123,167
Stratification of risk for thrombus formation and stroke	CT and Echo Doppler readings inform patient-specific vasculature and blood flow velocity	Multiple kernel learning to integrate morphological information, CFD simulations of haemodynamics and clinical record for risk stratification	Long-term monitoring of thrombosis and stroke risk because current inference and risk binning are static	33
Stratify risk for atherosclerotic plaque rupture and stroke	134 patient cohort with stenotic carotid reconstructed from MRI	ML model integrating imaging data, CFD simulations and other risk factors	Long-term monitoring of plaque rupture risk, infer treatment strategy for stenosis	32
Circulating tumour cell identification and classification	Prostate cancer (PC3) and breast cancer (SKBR3) cell trajectories in a microfluidic device inform cell parameters	LBM and IBM for FSI simulation, CNN and RNN for cell type classification	Integration of patient-specific tumour cell properties, validation of cell type classification against clinical data	74
Longitudinal haemodynamics of 3D blood flow (6 weeks) informed by wearable device	Inform coronary anatomy from angiogram, continuous wearable-derived heart rate and ECG; clinical measure of arterial pressure, cardiac output and haematocrit	LBM, longitudinal hemodynamic mapping framework with clustering	Cardiac event monitoring, inform clinical decisions and treatment planning	162
Vascular stenting and angioplasty surgical design for the peripheral pulmonary artery	Reconstruct patient-specific vascular and stenosis structure, inform CFD boundary condition with Womersley velocity profile	Proposed real-time intervention planning that integrates offline training with SimVascular CFD simulations and online probabilistic prediction for surgery	Continuous data integration, enrich model framework and training dataset to handle more complex lesion cases, improve user interaction	187
Vascular stenting surgical design for the coronary artery	Reconstruct patient-specific vascular structure based on angiography (n=73); inform CFD simulations with clinical measurement of cardiac output, cuff pressure, heart rate and haematocrit	One-dimensional blood flow simulation, ML regressors for real-time interpolation of post-intervention haemodynamics, extended reality front-end-integrated Harvis for interactive stenting design	Continuous data integration, enrich model framework and training dataset to handle more complex lesion cases	36
In silico trial for new lipid-reducing drug	Create a virtual-patient population representing atherosclerotic disease; calibrated with known physiology and previous clinical trial data	Probe virtual patients with representative age, sex, risk factor and lipid infiltration rate; ODE for drug pharmacokinetics	More comprehensive assessment for long-term efficacy and side effects, validation with actual patient population	191
Optimal treatment scheduling for patients with breast cancer	Dynamic contrast-enhanced and diffusion-weighted MRI images of ten patients with breast cancer to inform vessel geometry, vascular properties and tissue diffusion coefficient	One-dimensional fluid dynamics model for blood flow, 3D advection–diffusion model for drug distribution	Extend model prediction for drug distribution on tumour growth, validate optimized treatment plan	189

APR, adaptive physics refinement; CFD, computational fluid mechanics; CNN, convolutional neural network; CT, computed tomography; DNN, deep neural network; DPD, dissipative particle dynamics; ECG, electrocardiogram; FEM, finite-element method; FSI, fluid–structure interaction; IBM, immersed boundary method; LBM, lattice Boltzmann method; ML, machine learning; MRI, magnetic resonance imaging; ODE, ordinary differential equation; RNN, recurrent neural network.

haemodynamics under different disease conditions. For example, 3D cardiovascular models at the region of interest coupled with computationally efficient 0D models that inform realistic boundary conditions can achieve physiologically consistent predictions in patients[152,153]. Furthermore, digital models using lattice Boltzmann simulations can non-invasively estimate pulmonary pressures in patients with heart failure, with results validated against implantable monitors[154]. In addition, digital models incorporating short-term autoregulation through dynamic 1D representation with compliance feedback can provide a more comprehensive landscape of circulatory transport[155]. For example, lumped parameter models have been used in cerebral autoregulation investigations to understand ischaemic conditions[156,157], and regulation mechanisms in coronary arteries have been investigated by fluid modelling[158].

Although physics-based solvers remain the backbone of circulatory digital twins, data-driven modelling can rapidly approximate or improve the prediction of circulatory dynamics, particularly if repeated simulation or real-time evaluation is desirable. For example, large-scale parametric data have been used to train surrogate models that link geometric and flow features to pressure gradients and wall shear stress, enabling fast, minimal-data haemodynamic predictions[111]. Hybrid image-based and graph-based neural networks can reconstruct flow fields directly from patient-specific magnetic resonance images, achieving near-CFD accuracy while providing orders-of-magnitude faster inference[159]. Trained against CFD-derived ground truth, the model achieved 10% error while offering 5,000-times faster inference than CFD. In addition to patient-reconstructed vasculature, synthetic geometries can expand training diversity and improve model prediction. For example, a synthetic cohort of 1,000 aortic geometries, generated from statistical shape models, substantially improved velocity field prediction with errors below 5% (ref. 160). Similarly, recurrent architectures, trained on 8,516 virtual-patient datasets, enabled non-invasive estimation of arterial pressure waveforms in the carotid, femoral and brachial arteries[161].

Beyond static snapshots of digital models, digital shadows can capture longitudinal circulatory dynamics as they unfold over time, offering a more complete and dynamic representation of circulatory transport. In particular, long-term disease monitoring requires models that faithfully reflect patient-specific physiology across long timescales and incorporate continuous clinical data streams. This dynamic integration not only allows tracking of disease progression in real time but also reveals temporal patterns and cumulative changes in circulation that are otherwise inaccessible in single-time snapshot measurements. For example, a digital shadow can capture 3D haemodynamics over 4.5 million heartbeats by integrating continuous data from wearable devices[162]. This longitudinal haemodynamic mapping framework for coronary digital twins demonstrated that single-cycle haemodynamic metrics can considerably diverge from longitudinal maps, underscoring that cumulative exposure to persistently low wall shear stress provides more reliable indicators of disease-prone regions than static measurements.

Digital twins can further incorporate cardiovascular and cell modelling. For example, a multicomponent, multiphysics model, integrating the Alya cardiac electromechanics solver with the fluid dynamics solver HemeLB, couples 3D cardiac and blood transport simulations[163]. The coupled model indicates distinct muscle displacement and predicts aortic wall shear, highlighting the importance of cardiac-vascular coupling. It also demonstrates clinical relevance by simulating how myocardial scarring alters downstream vascular flow. Furthermore, a GPU-accelerated digital shadow, integrating cardiac contraction, electrical system conduction and flow dynamics, can simulate cardiac behaviour at the organ scale[164]. Here, the combination of message passing interface-based parallelization and GPU acceleration delivered substantial speedups, achieving cardiac simulations with nearly 1 billion spatial degrees of freedom to be completed within hours. These digital heart simulations align in range with cardiac metrics, such as ejection fraction and stroke volume, in a healthy population and in patients with left bundle branch block, and provided proof-of-concept predictions for cardiac resynchronization therapy in virtual patients, highlighting the potential of the digital heart for in silico clinical trials.

Digital shadows can also incorporate cell-level transport dynamics to capture the evolution of cell-related diseases. Ideally, these frameworks should resolve submicron cell–environment interactions across the metre-scale vasculature. However, direct simulation of trillions of red blood cells remains infeasible. Nevertheless, HPC and modelling algorithms, integrated with multiscale and multiphysics approaches, enable increasingly detailed and scalable physiological simulations. For example, alongside model parallelization and GPU acceleration, an adaptive physics refinement approach allows high-resolution modelling in regions of interest for cell tracking, while conserving computational resources with lower-resolution modelling in the remaining vascular geometry[165–167]. This approach enables circulating tumour cell tracking in conjunction with red blood cell transport in an upper body geometry, advancing the maximum modelling limit of accessible fluid volume by four orders of magnitude[123]. This multiscale integration is necessary for the recapitulation of cell physiology in the human circulatory system.

Moving beyond static representations, digital representations that capture system dynamics across time and space might enable digital twins that provide comprehensive and accurate representations of the human circulatory system.

Predicting disease outcome and risk stratification

A key application of digital twin-enabled bidirectional feedback is the prediction of disease outcomes and guidance of risk stratification. In particular, digital twins can aid in the prediction of diagnostic metrics that typically require invasive measurement, such as fractional flow rate. For example, a CFD-based digital model can accurately predict the fractional flow rate of coronary artery stenoses and identified longitudinal vorticity as a phenotype marker, which is not accessible for direct measurement[168]. Comparing the performance of fractional flow rate prediction in coronary artery disease across 23 CFD-based and 18 machine learning–based studies[108], both approaches achieve comparable areas under the receiver operating characteristic curve, with CFD-based methods demonstrating higher sensitivity but similar specificity relative to machine learning models. Importantly, CFD software platforms for non-invasive fractional flow rate estimation, including HeartFlow (CT-derived fractional flow rate) and CathWorks (angiography-derived fractional flow rate), have received approval by the US FDA.

In addition to predicting individual biomarkers, digital twins can aid in the risk stratification for vascular stenoses, aneurysms and hypertensive conditions. For example, a multimodal approach was applied to predict the risk of ischaemic stroke in patients with atrial fibrillation, integrating CT-derived left atrial morphology, CFD-simulated haemodynamics and electronic health records in a 130-patient cohort, stratifying the risk into three phenogroups with distinct profiles[33]. Similarly, risk stratification models were constructed for 343 patients with heart failure, combining haemodynamic measures, comorbidities, medications, blood tests and risk scores within unsupervised machine learning to identify phenogroups associated with an elevated risk of cardiovascular mortality[169]. By integrating AI with physics-based models, these hybrid approaches generate representative but inherently static digital twin frameworks that embed patient profiles at diagnostic time points to perform risk assessment. Their primary limitation is the absence of continuous data assimilation needed to adapt diagnoses longitudinally.

Digital platforms can also complement microfluidic systems designed to probe complex cellular behaviours. Microfluidic systems are often constrained by parameters that cannot be directly measured experimentally. Digital twins can bridge these gaps by expanding the set of accessible variables and providing mechanistic insights that microfluidics alone cannot capture. Thereby, digital twins can aid in disease diagnosis through in silico characterization of the mechanical properties of red blood cells and circulating tumour cells, such as stiffness, which typically differ between healthy and diseased states. For example, a microfluidic digital twin integrated with experimental measurements can quantify the shear modulus of red blood cells, identifying a potential marker for disease-associated alterations in cell mechanics[170]. Similarly, stiffness changes can be assessed during breast cancer progression by coupling microfluidics with computational modelling[171]. In addition, a foundational digital twin of the mechano–node–pore sensing platform can improve mechanophenotyping of single cells within subpopulations and expand the experimental capabilities of this device[172].

Digital twins can further be applied to trajectory-based sorting and computational isolation of circulating tumour cells. Circulating tumour cell abundance, which strongly correlates with metastatic risk, can be assessed using microfluidic devices combined with digital twin modelling to isolate circulating tumour cells from whole blood based on biomechanical traits[173,174]. For example, a hyperuniform micropore microfluidic device can differentiate prostate (PC3) and breast (SKBR3) cancer cells based on their migration trajectories[74]. Subsequently, these trajectories are incorporated into a fluid–structure interaction model to generate synthetic circulating tumour cell datasets, allowing convolutional and recurrent neural network classifiers that achieve 84% subtype classification accuracy.

Therefore, digital twins can shift disease risk assessment from population-based models to individualized prediction. By integrating mechanistic simulations with multimodal clinical data, digital twins can uncover hidden phenotypes, stratify patients by disease trajectory and identify those most likely to benefit from targeted interventions. Thereby, digital twins have the potential to become core components of precision medicine, enabling early diagnosis, accurate prognosis and proactive long-term management of diseases.

Personalized surgical planning

Digital twins predicting patient-specific haemodynamics provide tools for cardiovascular procedural and surgical planning, enabling clinicians to virtually test interventions and assess how altered flow patterns influence outcomes. By forecasting haemodynamic changes and vascular remodelling, these models help identify optimal treatment strategies and anticipate complications, such as restenosis, thrombosis or recurrence of arrhythmias, ultimately improving procedural precision and long-term patient outcomes. In particular, surgical vascular repair may trigger postoperative responses owing to long-term vascular remodelling, elevating the risk of restenosis and thrombosis, which can be assessed by digital models. Similarly, digital models can predict the impact of key haemodynamic metrics, such as wall shear stress, on vascular remodelling after stenting procedures[54,132]. These remodelling risks constitute a major aspect of surgical outcome assessment, serving as key evaluation to help with optimal surgical procedural planning[175–177].

By allowing patient-specific anatomical reconstruction, virtual device deployment and haemodynamic evaluation before intervention, digital twins can provide a personalized framework to anticipate surgical outcomes[65,66]. For example, computational models built off 0D–3D physics-based frameworks can predict surgical planning outcomes in congenital heart disease[178], post-hepatectomy[179,180] and pulmonary endarterectomy[181]. In addition, hybrid machine learning–CFD frameworks, using synthetic haemodynamic data to predict key metrics, such as instantaneous wave-free ratio and fractional flow reserve, in coronary artery bifurcation lesions, can support personalized pre-procedure prediction for coronary stenting procedures[182–184].

Digital twins that explicitly model surgical processes can perform iterative optimization of the surgical plan, considerably improving procedural success rates. For example, a personalized digital twin-assisted stenting workflow has been developed to treat pulmonary artery stenosis in congenital heart defects[185]. The workflow combines patient-specific anatomical reconstruction, simulation of stent and vessel expansion with foreshortening effects, and CFD analysis incorporating pulsatile inflow, Windkessel outflows and deformable vessel walls. The type, size and positioning of the stents were optimized according to the resulting haemodynamics in 16 patients, and the validation was performed using retrospective angiographic imaging and pressure measurements. Moreover, HarVI, a real-time digital twin for planning coronary interventions[36], integrates machine learning for rapid haemodynamic analysis with extended reality for intuitive 3D interaction with the design of the stenting procedure. Using one-shot training from 1D steady-state CFD simulations, the digital twin accurately reproduced key biomarkers, such as fractional flow rate, flow rate and wall shear stress, in a cohort of 73 patients. The workflow supports virtual treatment planning in 62 min, nearly ten times faster than conventional CFD-based approaches.

Digital twins can also aid in material optimization for grafting. For example, virtual CFD-based surgery has been performed to optimize pulmonary artery plasty by minimizing energy loss and wall shear stress in paediatric patients[186]. A digital twin of the thoracic aorta was used to design a prosthesis that is a more compliant than Dacron grafts for thoracic aortic aneurysm[53]. Here, a fluid–structure interaction analysis is incorporated to predict the haemodynamic outcomes of prostheses with different distensibility. Alternatively, a probabilistic neural twin can be applied for vascular repair in peripheral pulmonary artery stenosis[187]. This approach combines an offline workflow of patient-specific 3D modelling, reduced-order 0D approximations and iterative resistance tuning to simulate pretreatment and post-treatment haemodynamics. A probabilistic framework is then established to model treatment outcomes, followed by the development of an online neural twin trained on synthetic data to rapidly estimate joint and marginal haemodynamic probabilities, supporting real-time treatment planning.

Therefore, digital twins can improve cardiovascular surgery by enabling patient-specific predictive planning. Linking mechanistic modelling with clinical imaging and machine learning, they not only provide insight into surgical outcomes but also allow surgeons to virtually evaluate multiple strategies before entering the operating room. However, such digital twins would benefit from faster inference for real-time decision-making, greater design flexibility for surgical planning, more intuitive user interfaces for surgical workflow as well as standardized and comprehensive frameworks for surgical risk assessment and patient evaluation.

Optimal treatment planning

Digital twins can be applied for treatment planning by integrating systems pharmacology with patient-specific anatomy, molecular profiles and haemodynamics. These multiscale models can simulate drug behaviour in the context of individual vascular function, remodelling and flow patterns. For example, radiotherapy and chemotherapy rely on adequate blood flow for effective drug delivery and profoundly affect the circulatory system, causing collateral damage to healthy blood cells and vascular tissues. To prevent radiation-induced lymphopenia, the HEDOS framework can optimize radiation therapy doses to circulating blood cells on a whole-body flow scale[188]; here, a discrete-time Markov process is applied to simulate the spatiotemporal distribution of blood particles across the circulation and organs. Similarly, an advection–diffusion-based 3D digital shadow could optimize neoadjuvant doxorubicin treatment for ten patients with breast cancer[189], by using contrast-enhanced and diffusion-weighted MRI to estimate haemodynamic properties that inform drug delivery in individual patients. The optimized dose was chosen to balance the efficacy and toxicity of the treatment, markedly outperforming the standard protocol.

Digital twins can be further informed by lab-on-chip systems and mouse models that provide experimental measurements of the tumour microenvironment. For example, a breast cancer tumouroid model, made from patient-derived tumour cells, can be applied to investigate and amplify drug delivery by allowing the analysis of biomechanical determinants, such as vascular permeability and perfusion dynamics[190]. In addition, the REANIMATE pipeline can model drug uptake and treatment response on the cellular level by integrating longitudinal in vivo and ex vivo tumour imaging and 3D CFD-based simulations[171]. This pipeline revealed vascular and interstitial drug transport in murine tumour models and inferred drug response from a tumour vasculature-disrupting drug.

In addition to optimizing existing therapies, digital twins enable in silico clinical trials for the evaluation of new therapeutics in virtual-patient populations. For example, in the SIRIUS program, a quantitative systems pharmacology model was used to evaluate the lipid-lowering drug inclisiran in a virtual cohort of patients with atherosclerotic vascular disease[191]. By integrating lipoprotein metabolism, plaque dynamics and drug pharmacokinetics and pharmacodynamics, the trial simulated 5-year outcomes, including major cardiovascular events and mortality. Such frameworks allow the mechanistic evaluation of long-term drug effects without the constraints of real-world follow-up, offering a scalable and predictive approach to therapeutic development and optimization.

By integrating drug pharmacokinetic and pharmacodynamic profiles into circulatory transport modelling, digital twins can achieve personalized and optimized treatment planning, reducing treatment toxicity and accelerating the evaluation of new therapeutics through virtual clinical trials.

Outlook

Digital models and early-stage digital twin applications have been developed for human circulatory transport; however, next-generation digital twins must extend beyond static predictions to achieve continuous monitoring, adaptive model updating and long-term patient-specific disease forecasting. Realizing this capability will require overcoming fundamental barriers in data acquisition, physiological integration, model fidelity, computational efficiency and clinical translation (Box 1).

Fit-for-purpose modelling

An appropriate modelling scope must be defined that is aligned with the specific clinical or scientific question. Rather than defaulting to complex or high-fidelity formulations, model design should begin with identifying the minimum set of information required for the intended application[2]. This 'fit-for-purpose' consideration spans both spatial and temporal scales of modelling. Digital twin frameworks should be tailored to the dominant mechanisms of the target disease, avoiding

Box 1 | Translational considerations for digital twins in clinical use

Digital twin technology enables precision medical care to monitor individual patients and tailor interventions in real time. However, translating digital twins into clinical practice faces barriers beyond technical hurdles[15]. Key considerations include robust data management, strict data privacy, data security, ethical and equity issues, and regulatory and clinical integration hurdles, all of which must be addressed to ensure that digital heart or circulatory twins are safe, effective and acceptable in real-world health-care settings.

Integration of high-fidelity, multimodal health data is a major translational challenge owing to the lack of common data standards and interoperability protocols[4]. Data quality and completeness in the collection process are crucial for digital twins to generate the correct inference, because small deviations in the continuous data monitoring process could lead to false or misleading predictions[4]. Therefore, clinical translation will require robust data management strategies for multisource data integration frameworks, interoperability standards across health-care providers, rigorous data cleaning and population bias mitigation, ensuring that digital circulatory models are accurate and generalizable[196].

Reliability is also key to clinical adoption. Passive digital twin models that assist in treatment planning may lead the way in clinical application, whereas active and real-time digital twins that could impose direct influence on patients face additional regulatory and legal complexities. Physicians and surgeons might hesitate to fully adopt digital twins in daily practice, in particular, if artificial intelligence-driven suggestions conflict with their judgement[197]. Therefore, transparency and interpretability are essential in building trust, keeping the human in the loop in final decision-making.

In clinical use, data privacy and ethics need to be considered. Ethical frameworks typically demand that patients have rights over their health record; however, data privacy and ownership with regards to digital twins fall into a grey zone[198]. Companies or hospitals that use digital twins would have access to real-time patient health data, which requires additional legislation and regulation considerations to be compliant with patient rights. Digital twin initiatives must prioritize patient consent and limit data sharing to authorized, beneficial uses, with robust governance to prevent exploitation of data for unrelated commercial purposes[199].

unnecessary characterization of unrelated physiology. For circulatory transport, this might mean moving beyond static risk estimates to the modelling of how disease processes evolve over months to years. Greater attention should be dedicated to characterizing disease progression, such as short-term autoregulation and long-term vascular remodelling, to provide an accurate and dynamic representation for individual patients[51,52].

In addition, the trade-off between physical fidelity and computational latency must be navigated. High-fidelity models explicitly resolve multiscale physics but may not meet the real-time requirements of time-critical clinical workflows. For example, intraoperative or on-the-fly surgical planning may demand rapid model responses suited to surrogate approaches, such as reduced-order models or AI-based predictors[36,187]. The choice between maintaining full physical detail and adopting surrogate approximations should therefore reflect the primary objective of the application, whether that is interpretability, speed or scalability.

The diversity of clinical contexts and computational constraints might demand such a dynamic 'fit-for-purpose' strategy. Current digital models and digital shadows provide foundational components for integrating complex physiological systems. However, an adaptive framework, in which model scope and resolution evolve with data availability, computational resources and clinical needs, ensures that biomedical digital twins are both scientifically rigorous and operationally viable across applications.

Integration of AI and physics-based models

AI and physics-based modelling can be combined to produce biomedical predictions that are both computationally efficient and scientifically reliable. Physics-based solvers remain the backbone of mechanistic fidelity, capturing conservation laws, nonlinear flow behaviour and patient-specific anatomical detail. However, these models are computationally demanding, particularly for patient-specific or continuously evolving physiological conditions. AI can complement such solvers by inferring latent parameters, estimating dynamic boundary conditions from wearable or imaging data, and providing rapid approximations when full simulations are infeasible. Hybrid frameworks that fuse AI-driven inference with physics-based constraints offer a promising route towards digital twins capable of adjusting to real-time physiological changes while maintaining physical fidelity[15].

Within this integration, cardiovascular foundation models represent a long-term opportunity. Large pretrained neural operators or transformer-based PDE models, which are commonly applied in fluid dynamics, could provide generalizable priors for cardiovascular simulations[115,119–121]. However, cardiovascular-specific foundation models remain limited by data scarcity and the need for strict guarantees of physical accuracy and clinical reliability. Progress will require assembling multimodal datasets, embedding physical laws within AI architectures and defining clinically meaningful evaluation standards. Advancing these hybrid physics–AI methodologies will be central to enabling continuously adaptive digital twins for real-time monitoring, disease forecasting and individualized treatment planning.

Model fidelity and fairness

Verification, validation and uncertainty quantification are essential to establish trust in digital twins of circulatory transport. Verification ensures numerical correctness, whereas validation aligns model predictions with experimental or clinical observations[14]. Uncertainty quantification characterizes interindividual variability across genetics, physiology and disease states, and provides probabilistic estimates of confidence in model outputs. In biomedical applications, uncertainty must be handled with great caution, because model predictions can directly influence patient diagnosis or therapeutic decisions[1]. For patient safety, both the minimum and maximum bounds of uncertainty are important: underestimation of physiological responses can obscure potential complications, whereas overestimation can lead to unnecessary interventions. Thus, digital twins must move beyond mean accuracy and instead define clinically meaningful uncertainty ranges that demarcate safe operating thresholds[25]. Incorporating probabilistic risk metrics and sensitivity analyses can help identify parameters that disproportionately affect safety-critical outcomes, ensuring that decisions remain within acceptable risk limits even in the presence of uncertainty.

HPC resources

As digital twins evolve towards integrating complex physiological domains, HPC resources that efficiently distribute, parallelize and synchronize large-scale computations will be essential for maintaining model accuracy and scalability[125]. HPC architectures and workflows could benefit both model-centric and data-centric paradigms[5]. Model-centric applications might particularly benefit from advances in exascale computing and heterogeneous GPU–CPU architectures to enable higher-resolution and longer-term simulations. For data-centric approaches, HPC platforms could accelerate large-scale AI model training and data assimilation, allowing the fusion of high-throughput clinical datasets with physics-based inference in near-real time.

Cloud-based and hybrid computing infrastructures are also becoming key for the clinical translation of digital twin technologies. For example, hyperscalers, such as Amazon Web Services, Google Cloud Platform and Microsoft Azure, provide globally distributed data centres, automated scaling and specialized AI accelerators that allow hospitals and research institutions to execute computationally intensive digital twin workloads without maintaining large in-house clusters. These infrastructures offer managed compute resources and direct integration with health-data pipelines, enabling continuous model updates and rapid simulation turnaround. To accelerate deployment, digital twin research should prioritize collaborations that bridge hyperscaler ecosystems with HPC centres, creating interoperable workflows for high-fidelity clinical applications. Federal initiatives, such as the Genesis Mission, could further strengthen this landscape by uniting national supercomputing resources, large-scale federal datasets and emerging AI and quantum technologies to build and refine digital twins.

Continuous high-quality clinical data acquisition

Obtaining high-quality continuous data is essential for digital twin personalization, because deviations of model input and parameterization owing to inaccurate measurement amplify prediction error and confound model uncertainty quantification[1,4]. However, current measurements often lack the spatial and temporal resolution to capture rapid haemodynamic changes and vascular dynamics. Although technologies for non-invasive continuous monitoring are being explored, most clinically relevant measurements remain invasive. Invasive methods provide precise data but are constrained to short procedural windows and carry procedural risks, whereas non-invasive sensors enable daily monitoring but face issues such as signal noise, calibration drift and limited access to deep-vessel dynamics[192]. Advances in sensor biocompatibility and flexibility, noise reduction and software development are crucial in enabling accurate continuous monitoring of macrocirculation and microcirculation, providing the basis for a real-time digital twin of cardiovascular circulation[193].

Patient-specific personalization and feedback

Patient-specific personalization and closed-loop feedback are central priorities for biomedical digital twin research. A major constraint in personalization is parameterization: many physiological parameters required for accurate modelling are only indirectly observable. Accordingly, most digital models and digital shadows rely on population-average boundary conditions or generic cellular properties, introducing bias into patient-specific predictions. These limitations might be overcome by combining multimodal data acquisition with inference techniques, such as Bayesian calibration and machine-learning-assisted parameter estimation, to support robust data assimilation and individualized parameter tuning.

Designing feedback and intervention strategies that are intrinsically high-dimensional also remains a key challenge[2]. Current clinical practice in diagnosis, treatment planning and medication scheduling largely adheres to a one-size-fits-all paradigm. Therapeutic decisions are often guided by population averages or standardized guidelines, with limited capacity to account for interpatient variability in disease trajectories and treatment response[5,14]. However, clinical choices, such as stent size, shape, placement location or biomaterial, span a vast design space, with each decision interacting with patient-specific anatomy, disease state and long-term physiological adaptation. Digital twins could help navigate this space by integrating physics-based simulation with reinforcement learning, surrogate optimization and uncertainty-aware decision frameworks, to efficiently evaluate large sets of potential interventions.

Next-generation digital twins will increasingly assimilate patient data, update internal parameters as physiology evolves and deliver real-time guidance to support clinical decision-making. Although substantial scientific, computational and infrastructural challenges remain, advancing towards this vision promises to transition health care from reactive to predictive, from population-based guidelines to individualized strategies, and from episodic assessments to truly continuous, model-driven precision medicine.

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